

Assignment 2 Design Document

Design Checklist

1. What is the abstract thing you are trying to represent? Often the answer will be in terms of sets, sequences, and finite maps.

- **In this assignment, the abstract thing we are trying to represent is a polymorphic 2D array named `Array2` that can contain elements of any type.**

2. What functions will you offer, and what are the contracts of that those functions must meet?

- **`new(width: usize, height: usize, initial_value: T) -> Self { }`**
 - Initializes a new `Array2` with a defined width and height, and sets all elements equal to `initial_value`
- **`from_row_major(width: usize, height: usize, data: Vec<T>) -> Self { }`**
 - Constructs a new `Array2` from a vector in row major order
- **`from_col_major(width: usize, height: usize, data: Vec<T>) -> Self { }`**
 - Constructs a new `Array` from a vector in column major order
- **`iter_row_major(&self) -> Array2RowMajor<T>`**
 - Iterates through an `Array2` in row major order
- **`iter_col_major(&self) -> Array2ColMajor<T>`**

- Iterates through an Array2 in column major order
- **get(&self, width: usize, height: usize) -> Option<&T>**
 - Returns a reference to the element at the given coordinates

4. What representation will you use, and what invariants will it satisfy? (This question is especially important to answer precisely.)

- **Array2 will be represented with the values: width and height, as well as Vec<T> which is a 1 dimensional vector that contains the elements to be included in Array2.**

Invariants:

- Size of the Array2 should be equal to width * height which would be the same length as Vec<T>.
- When using the get() function to access an element at (x,y), x should be a value greater than or equal to 0 and less than width, and y should be a value greater than or equal to 0 and less than height .
- This method for designing our program allows us to represent a 2D array in the most straightforward way possible. Whether using the Row major or Column major order, the program can provide a method of iterating over the 2 dimensional array with a one to one representation of the elements within it corresponding to a specific x and y coordinate value.

We are confident enough in our design and the completeness of our answers that we decided that the questions on the design checklist (3, 5, 6, 7) would not provide any additional clarity to our thoughts. Thus, we completed checklist numbers 1, 2, and 4 as required.